Solution:

$$\overline{\overline{X}} = \frac{\sum \overline{X}}{10} = \frac{442}{10} = 44.2$$

$$\overline{R} = \frac{\sum R}{n} = \frac{58}{10} = 5.8$$

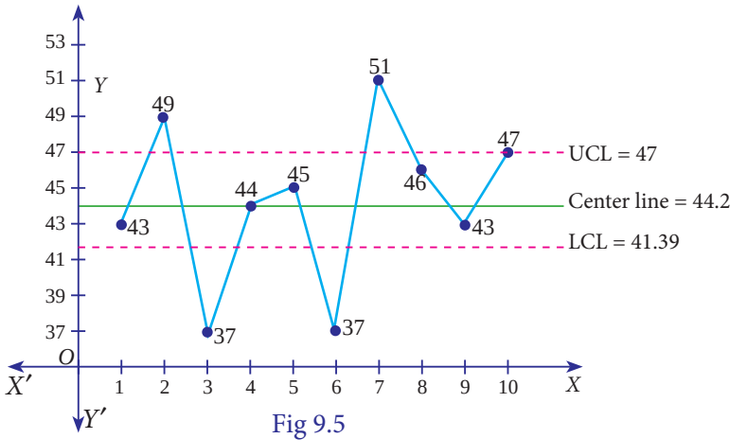


Fig 9.5

$$UCL = \overline{\overline{X}} + A_2\overline{R}$$
$$= 44.2 + 0.483(5.8) = 47.00$$
$$CL = \overline{\overline{X}} = 44.2$$
$$LCL = \overline{\overline{X}} - A_2\overline{R} = 44.2 - 0.483(5.8) = 41.39$$

The above diagram shows all the three control lines with the data points plotted, since four points falls out of the control limits, we can say that the process is out of control.

Exercise 9.3

1. Define Statistical Quality Control.
2. Mention the types of causes for variation in a production process.
3. Define Chance Cause.
4. Define Assignable Cause.
5. What do you mean by product control?
6. What do you mean by process control?
7. Define a control chart.
8. Name the control charts for variables.
9. Define mean chart.
10. Define R Chart.
11. What are the uses of statistical quality control?
12. Write the control limits for the mean chart.
13. Write the control limits for the R chart.
14. A machine is set to deliver packets of a given weight. Ten samples of size five each were recorded. Below are given relevant data:

Sample number	1	2	3	4	5	6	7	8	9	10
$\overline{X}$	15	17	15	18	17	14	18	15	17	16
R	7	7	4	9	8	7	12	4	11	5

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